

Linear combination and Linear Dependency

A vector V is called a **linear combination** of vectors $v_1, v_2, \dots, v_n$ if it can be expressed in the following form:

$$V = k_1 v_1 + k_2 v_2 + \dots + k_n v_n$$

$$\Rightarrow \begin{pmatrix} V_1 \\ \vdots \\ V_m \end{pmatrix} = k_1 \begin{pmatrix} v_{11} \\ \vdots \\ v_{1m} \end{pmatrix} + k_2 \begin{pmatrix} v_{21} \\ \vdots \\ v_{2m} \end{pmatrix} + \dots + k_n \begin{pmatrix} v_{n1} \\ \vdots \\ v_{nm} \end{pmatrix}$$

Here $k_1, k_2, \dots, k_n$ are scalars.

If $n = 1$, then the above equation will be reduced to $V = k_1 v_1$. It implies V is a linear combination of single vector v_1 if k_1 is a scalar multiple of v_1 .

Consider: $\begin{pmatrix} V_1 \\ \vdots \\ V_m \end{pmatrix} = k_1 \begin{pmatrix} v_{11} \\ \vdots \\ v_{1m} \end{pmatrix}$.

We evaluate the coefficients of $k_1, k_2, \dots$ where

$$V = k_1 v_1 + k_2 v_2 + k_3 v_3 + \dots$$

There exists no linear combination if there is no solution of the system (when the system is inconsistent).

Example: consider the vectors $v_1 = (2, 1, 4)$, $v_2 = (1, -1, 3)$, $v_3 = (3, 2, 5)$ in $\mathbb{R}^3$ (three dimensional vector space). Show that $V = (5, 9, 5)$ is a linear combination of v_1, v_2, v_3 .

Solution: In order to show that V is a linear combination of v_1, v_2, v_3 there must be scalars k_1, k_2, k_3 in a vector field such that

$$V = k_1 v_1 + k_2 v_2 + k_3 v_3$$

$$\Rightarrow (5, 9, 5) = k_1(2, 1, 4) + k_2(1, -1, 3) + k_3(3, 2, 5)$$

$$\Rightarrow \begin{pmatrix} 5 \\ 9 \\ 5 \end{pmatrix} = k_1 \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} + k_2 \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} + k_3 \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix}$$

$$\text{OR } (5, 9, 5) = (2k_1 + k_2 + 3k_3, k_1 - k_2 + 2k_3, 4k_1 + 3k_2 + 5k_3)$$

Equating corresponding components & formatting linear equations:

$$2k_1 + k_2 + 3k_3 = 5$$

$$k_1 - k_2 + 2k_3 = 9$$

$$4k_1 + 3k_2 + 5k_3 = 5$$

Augmented Matrix:

$$\begin{pmatrix} 2 & 1 & 3 & | & 5 \\ 1 & -1 & 2 & | & 9 \\ 4 & 3 & 5 & | & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & -1 & 2 & | & 9 \\ 2 & 1 & 3 & | & 5 \\ 4 & 3 & 5 & | & 5 \end{pmatrix} R_1, R_2 \text{ interchange}$$

$$= \begin{pmatrix} 1 & -1 & 2 & | & 9 \\ 0 & 3 & -1 & | & -13 \\ 0 & 7 & -3 & | & -31 \end{pmatrix} \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \end{matrix}$$

$$= \begin{pmatrix} 1 & -1 & 2 & | & 9 \\ 0 & 3 & -1 & | & -13 \\ 0 & 0 & -2 & | & -2 \end{pmatrix} R_3 \rightarrow 3R_3 - 7R_2$$

$$\begin{matrix} k_1 - k_2 + 2k_3 = 9 \\ 3k_2 - k_3 = -13 \\ -2k_3 = -2 \end{matrix}$$

By Back substitution we have: $k_3 = 1$, $k_2 = -4$, $k_1 = 3$

$$\therefore (5, 9, 5) = k_1(2, 1, 4) + k_2(1, -1, 3) + k_3(3, 2, 5)$$

$$\Rightarrow (5, 9, 5) = 3(2, 1, 4) - 4(1, -1, 3) + 1(3, 2, 5)$$

Hence (5,9,5) is a linear combination of (2,1,4), (1, -1, 3) and (3,2,5).